



MUHAMMAD SHOAIB

AI|ML & IMMERSIVE TECHNOLOGIES ENGINEER

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PROFILE INFO

Results-driven AI & Machine Learning and Immersive tech Engineer with 8+ years of experience designing and deploying production-grade ML systems. Specialized in deep learning, NLP, Computer Vision, Immersive Tech, and Edge Devices with a strong record of delivering measurable business impact. Adept at translating complex research into scalable solutions on cloud platforms, collaborating across teams, and mentoring junior engineers.

EDUCATION

University of engineering and technology, Lahore
(PhD Computer Science)
2025- Present

University of engineering and technology, Lahore
(MS Computer Science)
2019- 2022

COMSATS Institute of Information Technology
(BS Computer Science)
2014- 2018

Board of intermediate and secondary education, Faisalabad (Intermediate)
2010- 2012

Board of intermediate and secondary education, Faisalabad (Matriculation)
2008- 2010

WORK EXPERIENCE

Senior AI Consultant | National Center of AI (NCAI) – UET Lahore
Jan 2026 – Present • Lahore, Pakistan

- ▶ Contributing to national AI research across smart cities, healthcare diagnostics, and industrial automation.
- ▶ Advising on AI model architecture, deployment strategy, and research-to-product pipeline for NCAI projects.
- ▶ Collaborating with multi-disciplinary teams bridging academic research and real-world product delivery.

Development Team Lead – AI, ML & Spatial Computing | HawkLogix Pakistan
Aug 2023 – Present • Lahore, Pakistan

- ▶ Leading a cross-functional team of engineers in AI, ML, CV, NLP, AR, VR, and software development.
- ▶ Owning end-to-end project lifecycle: scoping, sprint planning, technical architecture, QA, and delivery.
- ▶ Designing and developing GenAI solutions using LLMs, Langchain and Vector Database.

Project Lead – AI & Emerging Tech | ProjectZ Incorporated
Jan 2023 – Oct 2025 • Lahore, Pakistan (Remote)

- ▶ Managed concurrent delivery of AI, software, and Immersive projects using Agile & Scrum methodologies.
- ▶ Ensured cross-team alignment on technical scope, timelines, and stakeholder communication.

CEO – EdTech Startup (PITB) | Punjab Information Technology Board
May 2023 – May 2024 • Lahore, Pakistan

- ▶ Founded and led CV and AR-based immersive learning startup under PITB's incubation program.

AWARDS & CERTIFICATIONS

C# Workshop CITT Sahiwal
Authority (COMSATS)
EXPO 2014

SEO Certification
Authority (WAPEXP)

WEB Certification
Authority (E Rozgaar)

Huawei ICT Competition
Middle East
(Got 2nd postion)
Authority (Huawei)

Robotics Summer Camp Best Teacher
Award
Authority (KICS, UET Lahore)

Spatial Computing Workshop
Authority (UMT)
International Conference 2025

SOFT SKILLS

- Leadership & Team Building
- Strategic Problem Solving
- Cross-Functional Collaboration
- Effective Communication
- Agile Project Management
- Analytical & Critical Thinking
- Adaptability & Innovation
- Mentorship & Knowledge Transfer

LANGUAGES

- English (Fluent)
- Urdu (Fluent)

WORK EXPERIENCE

Senior IT Consultant – AI & Immersive Tech | TatbeeqSoft
Sep 2022 – Dec 2023 • Lahore, Pakistan (Remote)
▸ Delivered Computer vision and Spatial computing solutions, integrating Raspberry Pi, Jetson Nano, and edge hardware for industrial clients.

Artificial Intelligence Engineer | YODO Dubai
Jan 2023 – Nov 2023 • Lahore, Pakistan
▸ Developed and deployed AI/ML models for simulation use cases, including RL-based game AI.

Team Lead AI | Alfa Mohuha
Aug 2022 – Jun 2023 • Lahore, Pakistan
▸ Led AI engineering team; oversaw model development, training pipelines, and production deployment.

Research Officer | University of Engineering & Technology (UET) Lahore
Oct 2017 – Jul 2022 • Lahore, Pakistan
▸ Conducted applied AI research in computer vision, NLP, and neural networks over nearly 5 years.
▸ Mentored junior researchers and contributed to institutional AI curriculum development.

AI & Python Programming Instructor | NAVTTC – at UET Lahore
Oct 2017 – Jul 2022 • Lahore, Pakistan
▸ Delivered hands-on training in Python, ML, deep learning, computer vision, and data analysis to 500+ students.

TECHNICAL SKILLS

ML / DL Frameworks
PyTorch • TensorFlow • Keras • Hugging Face Transformers • LangChain • Keras • scikit-learn • XGBoost • LightGBM

Languages & Technologies
Python • SQL • C# • C++ • JS • Bash • & Immersive, Web, Mobile Tech • Microservices

Cloud & MLOps
AWS • Azure ML • MLflow • Kubeflow • CI/CD Pipelines • FastAPI

Data Engineering
Pandas • NumPy • Feature Stores (Feast)

NLP & Vision
BERT • GPT • RAG • CLIP • YOLO • OpenCV • spaCy • Sentence Transformers • Detectron2 • SAM • MediaPipe • 3D Point Cloud

DevOps & Tools
Docker • Kubernetes • GitHub • GitHub Actions

REFERENCE

References will be provided upon request

CURRENT PROJECTS

MIRA (MEDICAL INTELLIGENT RECEPTIONIST AGENT) OVERVIEW:

MIRA IS A PRODUCTION-GRADE CONVERSATIONAL AI SYSTEM FOR AUTOMATED PATIENT CHECK-IN AND HEALTHCARE INTAKE WORKFLOWS USING LLMS, FASTAPI, AND MULTIMODAL INTERACTION PIPELINES.

CORE FEATURES:

- AI CONVERSATIONAL CHECK-IN ASSISTANT
- VOICE AND TEXT INTERACTION USING SST TTS AND VECTORLESS RAG
- PATIENT IDENTIFICATION WORKFLOWS
- INSURANCE VERIFICATION
- APPOINTMENT AND EHR INTEGRATIONS
- INSURANCE-CARD VISION PROCESSING
- STRUCTURED CONVERSATIONAL PARSING

TECHNOLOGIES: PYTHON • FASTAPI • CLAUDE • YOLO • TTS • STT • ELEVENLABS • DEEPGRAM • ASSEMBLYAI • LIVEAVATAR • UNITY ENGINE

PREVENTI AI OVERVIEW:

PREVENTI AI IS A SET OF BACKEND AI SERVICES FOR EXTRACTING STRUCTURED HEALTHCARE INFORMATION FROM PDFS, IMAGING REPORTS, LAB DOCUMENTS, AND FAX REFERRAL FORMS.

CORE FEATURES:

- INTELLIGENT MEDICAL DOCUMENT PARSING AND EXTRACTION
- PDF-BASED HEALTHCARE REPORT PROCESSING
- OCR-ASSISTED TEXT EXTRACTION FROM SCANNED MEDICAL DOCUMENTS
- PROCESSING OF LAB REPORTS, IMAGING SUMMARIES, AND FAX REFERRAL FORMS
- AI-GENERATED NARRATIVE SUMMARIES FROM MEDICAL FINDINGS
- LIFESTYLE AND PREVENTIVE HEALTHCARE RECOMMENDATIONS USING LLMS
- FASTAPI-BASED REST APIS FOR DOCUMENT INGESTION AND PROCESSING
- ENVIRONMENT-DRIVEN CONFIGURATION FOR SCALABLE DEPLOYMENT
- DOCKERIZED DEPLOYMENT FOR CLOUD ENVIRONMENTS
- CI/CD AUTOMATION USING GITHUB ACTIONS AND AZURE DEPLOYMENT WORKFLOWS
- STRUCTURED LOGGING AND CENTRALIZED REQUEST/RESPONSE HANDLING
- PRODUCTION-READY HEALTHCARE AI BACKEND SERVICES

TECHNOLOGIES: PYTHON • FASTAPI • PYMUPDF • RAPIDOCR • CLAUDE

IABETES 360 OVERVIEW:

PREVENTI AI IS AN AI-POWERED HEALTHCARE SUPPORT PLATFORM FOCUSED ON DIABETES MANAGEMENT, NUTRITION ASSISTANCE, AND PERSONALIZED HEALTH GUIDANCE. THE SYSTEM LEVERAGES MULTIMODAL AI CAPABILITIES TO ANALYZE USER INPUTS INCLUDING TEXT, VOICE, AND MEAL IMAGES TO PROVIDE INTELLIGENT RECOMMENDATIONS, GLUCOSE-RELATED INSIGHTS, NUTRITION GUIDANCE, AND CONVERSATIONAL SUPPORT THROUGH SCALABLE FASTAPI BACKEND SERVICES.

CORE FEATURES:

- AI-POWERED DIABETES SUPPORT ASSISTANT
- PERSONALIZED NUTRITION AND MEAL GUIDANCE
- MEAL-IMAGE ANALYSIS USING MULTIMODAL AI MODELS
- CONVERSATIONAL AI SUPPORT FOR HEALTH-RELATED QUERIES
- AI-GENERATED FOOD RECOMMENDATIONS
- HEALTH AND GLUCOSE INSIGHT GENERATION
- VOICE-INPUT SUPPORT WITH AUDIO TRANSCRIPTION
- FASTAPI-BASED BACKEND APIS WITH STRUCTURED REQUEST/RESPONSE MODELS
- LARGE LANGUAGE MODEL (LLM) INTEGRATION FOR INTELLIGENT RECOMMENDATIONS
- MULTIMODAL AI WORKFLOWS COMBINING TEXT, IMAGE, AND AUDIO INPUTS
- CI/CD DEPLOYMENT AUTOMATION PIPELINE
- PRODUCTION-READY HEALTHCARE AI BACKEND INFRASTRUCTURE

TECHNOLOGIES: PYTHON • FASTAPI • PYMUPDF • RAPIDOCR • CLAUDE

ROUNDING SCHEDULE OVERVIEW:

DEVELOPED AN INTELLIGENT PHYSICIAN SCHEDULING PLATFORM TO AUTOMATE FACILITY ROUNDING ASSIGNMENTS USING FORECASTING MODELS, WORKLOAD BALANCING, AND CONSTRAINT-BASED OPTIMIZATION.

- **SCHEDULE DAY INPUTS:** THE PROCESS BEGINS BY COLLECTING THE PHYSICIAN AND FACILITY SCHEDULE, PREVIOUS DAY'S FACILITY SCHEDULE, PHYSICIAN-FACILITY CREDENTIAL STATUS, AND 28 DAYS OF HISTORIC CASE COUNT DATA.
- **CASE COUNT ESTIMATION:** AN AI MODEL IS USED (IF AVAILABLE) TO FORECAST THE CASE COUNT; OTHERWISE, AN AVERAGE CASE COUNT IS USED — BOTH PATHS LEAD TO AN ESTIMATED CASE COUNT.
- **PHYSICIAN ASSIGNMENT ALGORITHM:** USES AVERAGE CASE TIME PER PHYSICIAN PER FACILITY AND ROUNDING DURATION TO FACILITIES AS CORE INPUTS FOR GENERATING ASSIGNMENTS.
- **FACILITY PRIORITIZATION:** FACILITIES WITH THE LEAST CREDENTIALLED PHYSICIANS ARE PRIORITIZED FIRST, AND THE PREVIOUS SCHEDULE'S PHYSICIAN IS REASSIGNED WHERE POSSIBLE.
- **NH PHYSICIAN ASSIGNMENT:** IF UNASSIGNED FACILITIES REMAIN AFTER REASSIGNMENT, A PRIORITY NH PHYSICIAN WITH THE LEAST WORKLOAD IS ASSIGNED; IF NO FACILITIES REMAIN UNASSIGNED, THE PROCESS ENDS.
- **STROKE PHYSICIAN ASSIGNMENT:** FOR ANY STILL-UNASSIGNED FACILITIES, THE SYSTEM COMPUTES STROKE PHYSICIAN WORKLOAD, FREE TIME, AND PRIORITY THEN ASSIGNS THE STROKE PHYSICIAN WITH THE LEAST WORKLOAD BEFORE CONCLUDING THE PROCESS.